

Instructions

Unless a question states otherwise, take $g = 9.8 \text{ m s}^{-2}$ wherever a numerical value of g is required. Solutions also give final answers for $g = 10 \text{ m s}^{-2}$ where these differ.

Questions

Question 1	<i>6 marks</i>	2
Question 2	<i>11 marks</i>	3
Question 3	<i>7 marks</i>	4
Question 4	<i>5 marks</i>	5
Question 5	<i>4 marks</i>	6
Question 6	<i>7 marks</i>	7
Question 7	<i>6 marks</i>	8
Question 8	<i>5 marks</i>	9
Question 9	<i>6 marks</i>	10

1. Three particles of masses m , $2m$ and km are placed at the points $A(a, 0)$, $B(-2a, 3a)$ and $C(2a, -a)$ respectively, where $k > 0$.

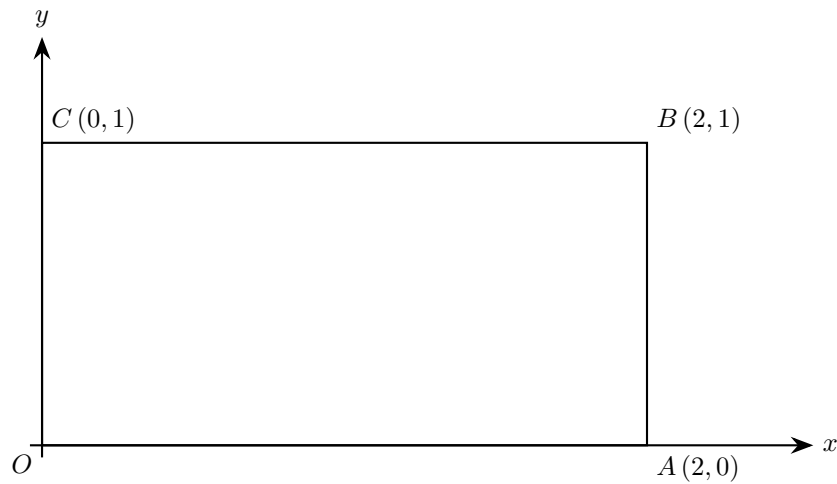
The centre of mass of the three particles is the point with coordinates $(\bar{x}, \bar{y})$.

- (a) (i) Find $\bar{x}$ in terms of a and k .
(ii) Find $\bar{y}$ in terms of a and k .

[4]

- (b) Given that the centre of mass lies on the circle with centre $(a, 0)$ and radius $\frac{\sqrt{2}}{2}a$, find the possible values of k .

[2]



2. The diagram shows a light rectangular lamina $OABC$ where O is the origin of the coordinate system in which the points A , B and C have coordinates $(2, 0)$, $(2, 1)$ and $(0, 1)$ respectively. Coordinates refer to the axes shown and the units are metres.

The following three forces are applied to the lamina.

$$\begin{pmatrix} p \\ 2q \end{pmatrix} \text{ N at the point } (1, 0)$$

$$\begin{pmatrix} -q \\ p \end{pmatrix} \text{ N at the point } C$$

$$\begin{pmatrix} -3 \\ -6 \end{pmatrix} \text{ N at the point } (t, 1), \text{ where } 0 \leq t \leq 2$$

The three forces form a couple of magnitude 3 N m in the anticlockwise direction.

- (a) Determine the values of p , q and t .

[5]

The three forces are removed. The lamina is in the same position as shown in the diagram.

Three particles are now attached to certain points on the lamina as follows.

- (b) Find the coordinates of the composite body's centre of mass. Give your answer in terms of λ , where the particles are
- a particle of mass 2 kg at the point $(2\lambda^2, \lambda)$, where $0 \leq \lambda \leq 1$
 - a particle of mass 1 kg at the point A
 - a particle of mass 1 kg at the point C

[3]

- (c) Show that, as λ varies, the centre of mass of the combined object lies on the curve

$$x = 4y^2 - 2y + k$$

where k is a constant to be determined.

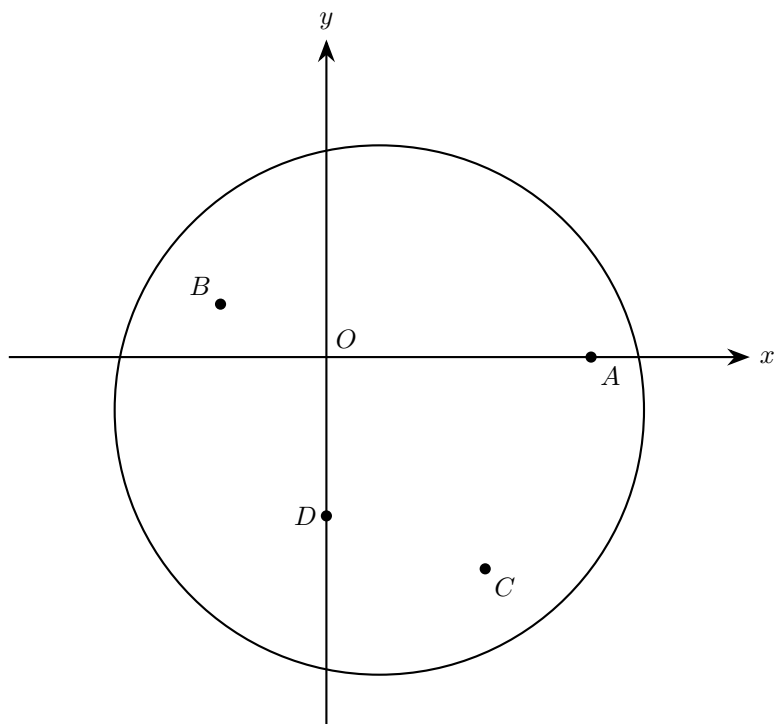
[3]

- 3.** Three particles of masses $2m$, $5m$ and $3m$ are placed at the points $(1, -4)$, $(-3, 2)$ and $(q, 2q - 1)$ respectively.

The value of q is such that the distance of the centre of mass of the three particles from the point $(1, 0)$ is as small as possible.

Find the value of q .

[7]



4. Four sensors are attached to a circular plate.

The plate is to be modelled as a uniform lamina and the four sensors as four particles.

The diagram shows the lamina, the four particles A , B , C and D , and the coordinate axes.

The lamina has mass 6 kg and its centre of mass is at the point $(1, -1)$.

Particle A has mass 4 kg and is at the point $(5, 0)$.

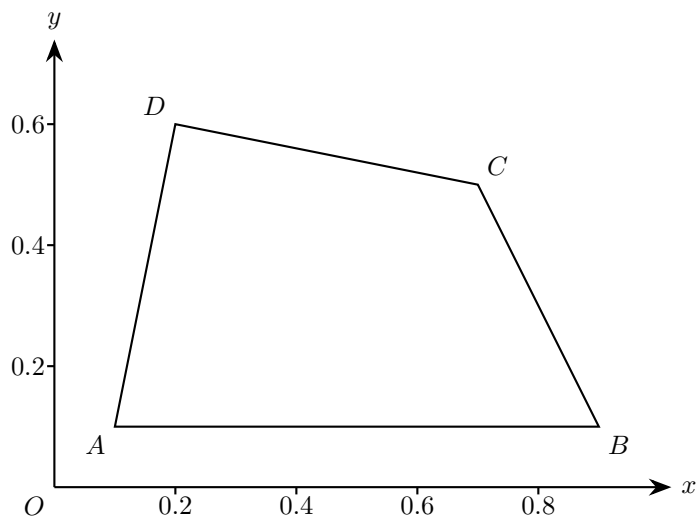
Particle B has mass 3 kg and is at the point $(-2, 1)$.

Particle C has mass 5 kg and is at the point $(3, -4)$.

Particle D has mass 2 kg and is at the point $(0, -3)$.

Find the coordinates of the centre of mass of the system of plate and sensors.

[5]



5. Particles A , B , C and D of masses 1 kg, 2 kg, 3 kg and 4 kg respectively are joined by light rigid rods to form the quadrilateral frame shown in the diagram. The particles are at the points $A(0.1, 0.1)$, $B(0.9, 0.1)$, $C(0.7, 0.5)$ and $D(0.2, 0.6)$, where coordinates are measured in metres.

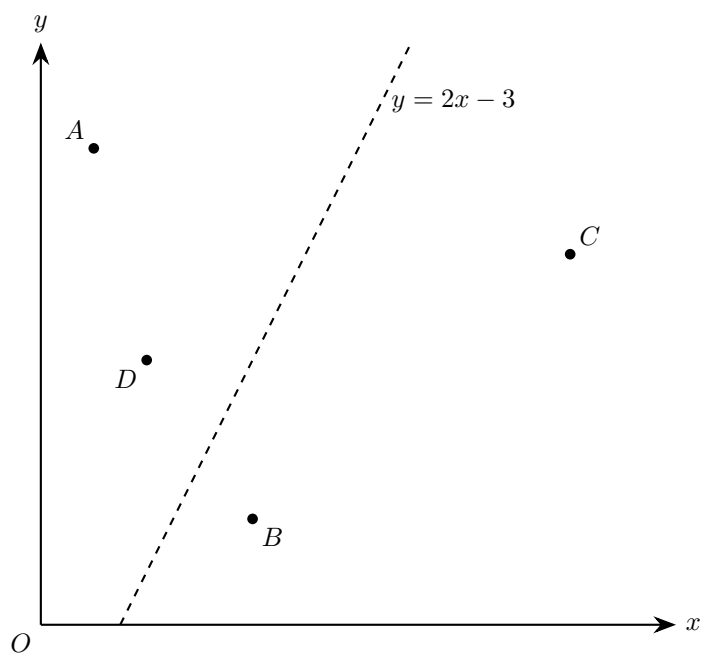
G , the centre of mass of the frame, has coordinates $(\bar{x}, \bar{y})$.

- (a) Show that $\bar{y} = 0.42$. Find $\bar{x}$. [3]
- (b) Using your answer to part (a), explain why the frame cannot be in equilibrium in a horizontal plane if it is supported by a single peg touching the frame at one point. [1]

6. Four particles of mass m , $2m$, $4m$ and λm are placed at points with coordinates $(0, 0)$, $(2, 3)$, $(1, -1)$ and $(4, 2)$ respectively. The centre of mass of the system is at $(2, k)$.

(a) Show that $\lambda = 3$. [4]

(b) Calculate the value of k . [3]



7. The diagram shows four particles, A , B , C and D , fixed in a horizontal plane which contains the x - and y -axes.

Particle A has mass 3 kg and is attached at the point $(1, 9)$.

Particle B has mass 4 kg and is attached at the point $(4, 2)$.

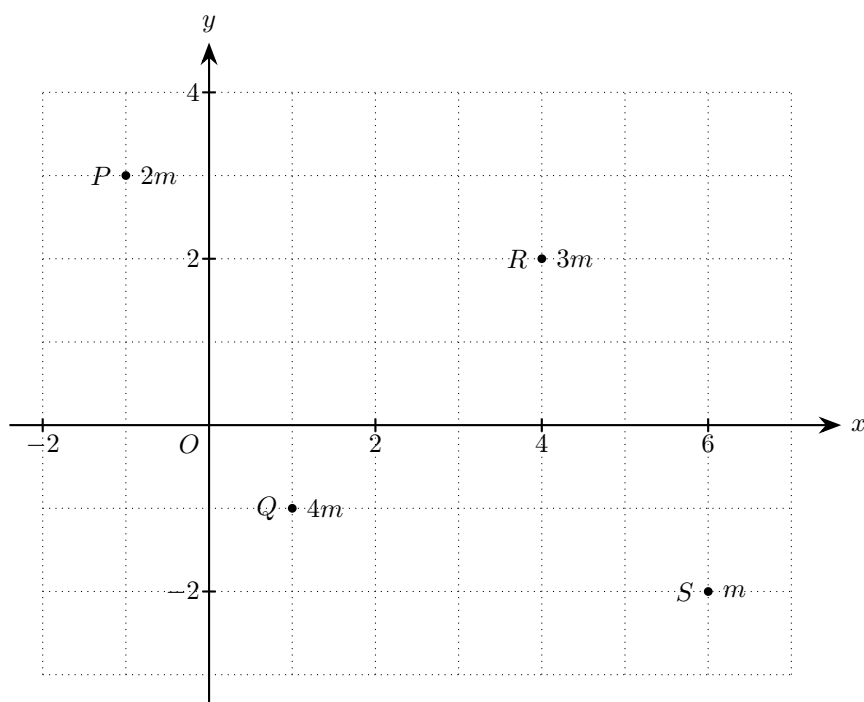
Particle C has mass 5 kg and is attached at the point $(10, 7)$.

Particle D has mass m kg and is attached at the point $(2, 5)$.

The centre of mass of the four particles lies on the line $y = 2x - 3$.

Find the value of m and the coordinates of the centre of mass.

[6]



8. The diagram shows a system of four particles of masses $2m$, $4m$, $3m$ and m situated in the x - y plane at the points $P(-1, 3)$, $Q(1, -1)$, $R(4, 2)$ and $S(6, -2)$ respectively.

Find the coordinates of the centre of mass of the system.

[5]

- 9.** Four particles of masses m , $2m$, $3m$ and km are positioned at the points with coordinates $(-a, a)$, $(2a, -2a)$, $(0, 0)$ and $(\mu a, 2\mu a)$ respectively, where k and μ are constants.

The centre of mass of the four particles is at the point with coordinates $(2a, 3a)$.

(a) Find the value of k . **[3]**

(b) Find the value of μ . **[3]**